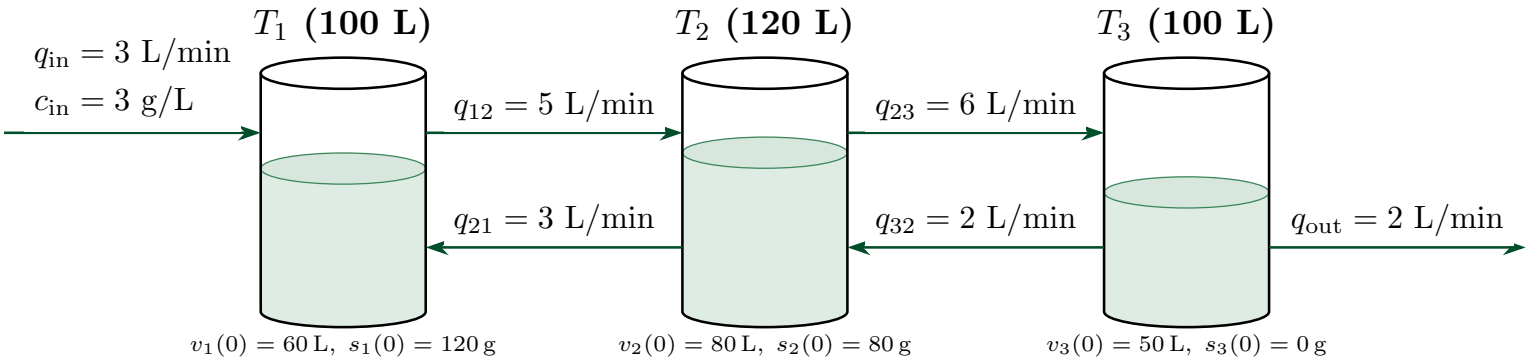
MA3521 — Well-Mixed Tanks

Name:

Problem 1

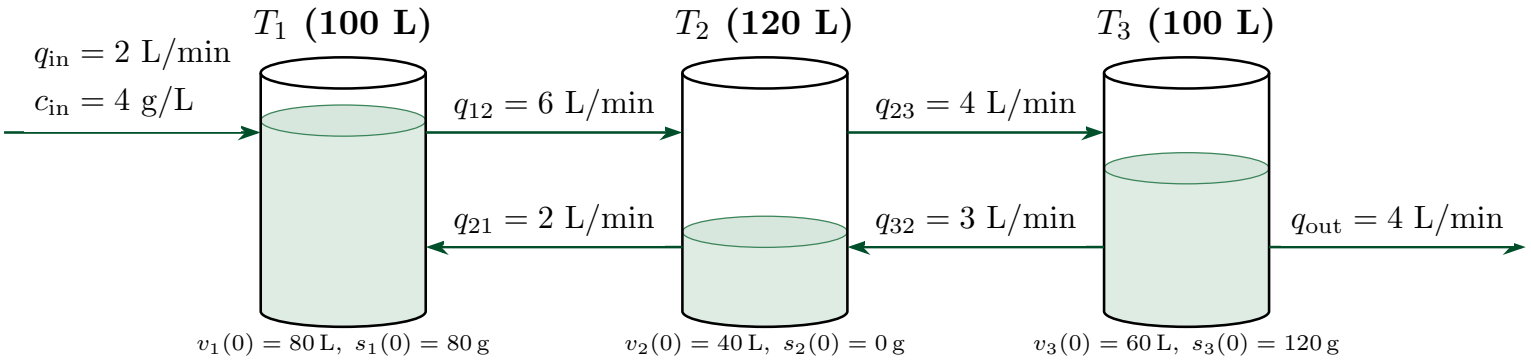


The IVP $\mathbf{s}' = A\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \mathbf{s}_0$ makes sense for $0 < t <$ ____ with

$$\mathbf{v}(t) = \begin{pmatrix} \\ \\ \\ \end{pmatrix} \quad \mathbf{s}_0 = \begin{pmatrix} \\ \\ \\ \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} \\ \\ \\ \end{pmatrix}$$

$$A = \begin{pmatrix} & & \\ & & \\ & & \\ & & \end{pmatrix}$$

Problem 2



The IVP $\mathbf{s}' = A\mathbf{s} + \mathbf{b}$, $\mathbf{s}(0) = \mathbf{s}_0$ makes sense for $0 < t <$ ☐ with

$$\mathbf{v}(t) = \begin{pmatrix} \\ \\ \\ \end{pmatrix} \quad \mathbf{s}_0 = \begin{pmatrix} \\ \\ \\ \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} \\ \\ \\ \end{pmatrix}$$

$$A = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$